

Gaurav Sahoo

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Education

Vellore Institute of Technology, Vellore

Sep 2022 – Jul 2027

- Integrated M.Tech in Computer Science – Data Science Specialization
- CGPA: 8.54

Modern Senior Secondary School, Chennai (CBSE)

Apr 2008 – Jul 2022

- 10th – 91% (CBSE)
- 12th – 92.5% (CBSE)

Technical Skills

- **Languages:** Python, SQL, Java, C/C++, R, JavaScript
- **Core Concepts:** Data Structures & Algorithms, OS, OOP, DBMS, Computer Networks, Software Engineering
- **Data Science & ML:** Scikit-learn, PyTorch (Hugging Face/SBERT), Statistical Modelling, Deep Learning
- **NLP & GenAI:** spaCy, NLP Pipelines, Named Entity Recognition, LLM/GenAI API Integration (Gemini), Prompt Engineering, RAG
- **MLOps & Data Engineering:** MLflow, Evidently AI, Docker, GitHub Actions CI/CD, ETL/ELT
- **Tools & Platforms:** Git/GitHub, Streamlit, PostgreSQL, MySQL, MongoDB, Linux (Ubuntu), VS Code, Jupyter, Google Colab, Power BI

Projects

MedScribe-IQ – Clinical Report Auditor & ICD-10 Validator

[GitHub](#)

Tech Stack: Python, spaCy, Gemini LLM, MLflow, Evidently AI, Streamlit, Docker

- Built a clinical text pipeline using **spaCy**, custom medical dictionaries, and negation detection.
- Added **ICD-10 validation** through the WHO API with a local fallback mechanism.
- Integrated **Gemini LLM** for audit summaries and tracked experiments with **MLflow**.
- Monitored data drift using **Evidently AI** and deployed the dashboard with **Streamlit + Docker**.

SkillMatch – AI Resume vs Job Fit Evaluator

[GitHub](#)

Tech Stack: Python, PyMuPDF, SBERT, spaCy, Gemini API, Streamlit, Docker

- Developed an AI resume evaluator using **PyMuPDF** for PDF parsing and Gemini API for job-role analysis.
- Implemented **SBERT (all-MiniLM-L6-v2)** for semantic matching between resumes and job descriptions.
- Built **spaCy**-based skill extraction and generated structured skill-gap recommendations.
- Deployed the application using **Streamlit** and containerized it with **Docker**.

Predictive AQI Monitoring & Visualization System

[GitHub](#)

Tech Stack: Python, Random Forest, XGBoost, Streamlit, Folium, Docker

- Built a real-time AQI monitoring and forecasting application using the **OpenWeatherMap API**.
- Developed interactive geospatial visualization using **Folium** with Geopy/Nominatim location search.
- Trained **Random Forest** and **XGBoost** models for session-based AQI forecasting.
- Served predictions through **Streamlit** and containerized the application with **Docker**.

Experience

Research Intern, CDMM, VIT

May 2025 - Jun 2025

- Built a real-time data analysis platform to track **Random Forest** and **XGBoost** models using MLflow.
- Managed the end-to-end pipeline from **data ingestion and experimentation to visualization**.

Data Science Intern, Missing Connectz Technology Solutions, Chennai

Jun 2025 - Jul 2025

- Worked with commercial datasets across data preparation and core **data analysis workflows**.
- Developed automated data pipelines and backend integrations using **Django**.

Finance Head, International Society of Automation (ISA), VIT Chapter

Jan 2024 - Dec 2024

- Managed chapter finances, budgeting, and resource allocation across student-led activities.
- Served as *Core Committee Member* (Jul 2023 – Jan 2024), coordinating cross-functional stakeholders.

Achievements & Certification

- **Certification:** Machine Learning A-Z: AI, Python & R [2025] — Udemy, Kirill Eremenko & Team
- **Languages:** English, Hindi, Odia, Tamil